National Research University Higher School of Economics (HSE)

Curriculum

Field of study 01.03.02 Applied Mathematics and Information Science

Educational Programme "Data Science and Business Analytics"

Specializations: "Data Analysis in Applied Research", "Data Science in Business", "Data Science in Finance"

Implementing unit: Faculty of Computer Science, HSE - Moscow

1 st, 2025/2026 academic year

APPROVED

06.06.2025

Vice Rector

ROSHCHIN S.Y.

Signed with EDS

Length of Programme: 4 years

Years of Study: 2025/2026 - 2028/2029

Mode of Study: Full Time

Degree: Bachelor's degree

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Progamme				**60,00**	**2 480**	**1 306**	**247**	**271**	**378**	**410**	
	Data Culture						**2**			**2**		
1	Independent Digital Literacy Test	C	отдел развития цифровых компетенций				2			2A		Online Course, Foreign language
	Major				**41,00**	**1 558**	**726**	**142**	**142**	**182**	**260**	
	Core Professional Block (Major)				**41,00**	**1 558**	**726**	**142**	**142**	**182**	**260**	
1	Introduction to Python for Data Science	C	Department of Big Data and Information Retrieval		4,00	152	60				60A	Foreign language
2	Algebra	C	Department of Big Data and Information Retrieval		3,00	114	40				40A	Foreign language
3	Algorithms and Data Structures 1	C	Department of Big Data and Information Retrieval		3,00	114	80				80A	Foreign language
4	Discrete Mathematics	C	Department of Big Data and Information Retrieval		7,00	266	102	30A	30	42A		Foreign language
5	Linear Algebra and Geometry	C	Department of Big Data and Information Retrieval		9,00	342	136	28	28A	40	40A	Foreign language
6	Calculus	C	Department of Big Data and Information Retrieval		9,00	342	136	28	28A	40	40A	Foreign language
7	C++ Programming Language	C	Department of Big Data and Information Retrieval		6,00	228	172	56A	56	60A		Foreign language
	General Courses				**10,00**	**580**	**304**	**49**	**73**	**114**	**68**	
	Optional General Courses				**9,00**	**342**	**142**	**12**	**36**	**70**	**24**	
1	Russian History	C	School of History		4,00	152	76	12	16A	24	24A	Online Course, Foreign language
2	Foundations of Russian Statehood	C	School of History		2,00	76	26			26A		Online Course, Foreign language
3	Economics	C	Department of Big Data and Information Retrieval		3,00	114	40		20	20A		Online Course, Foreign language
	Core General Courses				**1,00**	**238**	**162**	**37**	**37**	**44**	**44**	

1	Safe Living Basics	C	Проектно-учебная лаборатория «Бизнес-инкубатор Высшей школы экономики»		1,00	38	18	9	9A			Online Course, Foreign language
2	Physical Training	C	Department of Physical Training			200	144	28	28	44	44A	Foreign language
	Language and Communication				**9,00**	**342**	**274**	**56**	**56**	**80**	**82**	
	Examinations				**9,00**	**342**	**274**	**56**	**56**	**80**	**82**	
1	English Language	C	Department of Big Data and Information Retrieval		9,00	342	272	56	56A	80	80A	Foreign language
2	Independent English Exam	C	School of Foreign Languages				2				2A	Foreign language

Curriculum agreed:

Academic Supervisor	Voznesenskaia T.V.	23.05.2025
Dean	ARZHANTSEV I.V.	27.05.2025
Head of Centre for Educational Model Design	LEPESHKIN I.A.	06.06.2025

* Subject type:

Compulsory course	C
Elective course	E